

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
AMTI – NMTC – 19th October, 2024 – INTER FINAL

Instructions:

1. Answer all questions. Each question carries 10 marks.
 2. Elegant and innovative solutions will get extra marks.
 3. Diagram and justification should be given wherever necessary.
 4. Before start answering, fill in the FACE SLIP completely.
 5. Your 'rough work' should be in the answer sheet itself.
 6. Maximum time allowed is THREE hours.
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1. The numbers $1, 2, \dots, 9$ are randomly arranged on a circle. Prove that there are three adjacent numbers whose sum is at least 16.
2. Find all quadruples (a, b, c, d) of positive integers, where $a \geq b \geq c \geq d$ such that

$$\left(1 + \frac{1}{a}\right) \left(1 + \frac{1}{b}\right) \left(1 + \frac{1}{c}\right) \left(1 + \frac{1}{d}\right) = 5.$$

3. Let $P(x)$ be a polynomial with integer coefficients. If $P(101)P(97) = -16$, find the value of $P(101) + P(97)$.
4. Let $ABCD$ be a quadrilateral and let P be a point in its interior. Let K, L, M, N be the feet of the perpendiculars from P onto the lines AB, BC, CD, DA respectively. Let H_a, H_b, H_c, H_d be the orthocenters of the triangles AKN, BKL, CLM, DMN respectively. Prove that H_a, H_b, H_c, H_d are the vertices of a parallelogram.
5. Let M be the midpoint of the side BC of a triangle ABC . Point K on the segment AM satisfies $CK = AB$. Denote by L the intersection of CK and AB . Prove that the triangle AKL is isosceles.
6. If $a, b, c > 0$ such that $abc = 1$, prove that

$$\frac{1}{ab + a + 2} + \frac{1}{bc + b + 2} + \frac{1}{ca + c + 2} \leq \frac{3}{4}$$

7. P is a regular polygon with 2025 vertices. The vertices are colored with colors red, green and blue such that any two adjacent vertices have different colors. Show that we can divide the polygon into triangles with non intersecting diagonals whose end points have different colors.
8. During a break, 128 children at a school sit in a circle to play a game. The teacher walks clockwise around the circle to hand out chocolates to the children according to the following scheme: he selects one child and hands over a chocolate to the child, then he skips the next child and gives a chocolate to the next one, then he skips 2 children and gives a chocolate to the next one, then he skips 3 children and so on. Show that after many rounds, each child will have received at least one chocolate.
If there were 129 children to start with, will the same conclusion hold?